

OSSP:

Commands:

uname: Displays the operating system and kernel information.

lscpu: Displays processor details like CPU model, cores, and architecture.

```
shivam@MSI123:~$ nano shivam.c
shivam@MSI123:~$ gcc shivam.c -o shivam
shivam@MSI123:~$ ./shivam
```

Enter a Linux command: uname

Parent Process

Parent PID : 500

Child PID : 501

Child Process

Child PID : 501

Parent PID: 500

Executing command: uname

ps: Lists all active processes currently running in the system.

shivam@MSI123:~\$./shivam

Enter a Linux command: ps

Parent Process

Parent PID : 515

Child PID : 516

Child Process

Child PID : 516

Parent PID: 515

Executing command: ps

PID	TTY	TIME	CMD
335	pts/0	00:00:00	bash
515	pts/0	00:00:00	shivam
516	pts/0	00:00:00	ps

Relation Between Hardware Resources and Operating System Services:

1. CPU

The operating system manages CPU usage by sharing processor time among different programs.

Example: Running a browser and a media player at the same time.

2. Memory (RAM) -> Memory Management

The OS assigns RAM to applications and prevents them from accessing each other's memory.

Example: VS Code and Chrome use separate memory while running.

3. Storage (SSD) → File System Management

The OS manages files and folders, allowing users to save and access data easily.

Example: Saving a project file on the desktop.

4. I/O Devices → Device Management

The OS uses drivers to communicate with hardware devices and handle input/output operations.